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RESEARCH ARTICLE

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A multi-stakeholder preliminary needs assessment to inform the co-design of a digital telerehabilitation tool for children and adolescents with ADHD

Marina Rivas-García[a,b], Carmen Vidal-Ramírez[a], Marta Mora-Traverso[c], Abel Toledano-González[d], María del Carmen Rodríguez-Martínez[e], Esther Molina-Torres[f], José-Antonio Marín-Marín[g], José-Matías Triviño-Juárez[h], Miguel Gea-Mejías[i] and Dulce Romero-Ayuso[j,k,l]

[a]Faculty of Medicine, University of Granada, Granada, Spain; [b]Hospital Universitario San Cecilio, Granada, Spain; [c]University of Zaragoza, Zaragoza, Spain; [d]Department of Psychology, University of Castilla-La Mancha, Ciudad Real, Spain; [e]Departament of Physical Therapy, Occupational Therapy Division, Faculty of Health Sciences, University of Malaga, Málaga, Spain; [f]Specialized Educational Guidance Team of Granada, Department of Educational Development and Vocational Training, Andalusian Government, Granada, Spain; [g]Department of Didactics and School Organization, Faculty of Science Education, University of Granada, Granada, Spain; [h]Department of Radiology and Physical Medicine, Faculty of Medicine, University of Granada, Granada, Spain; [i]Department of Computer Languages and Systems, School of Computer and Telecommunications Engineering, University of Granada, Granada, Spain; [j]Department of Physical Therapy, Occupational Therapy Division, Faculty of Health Sciences, University of Granada, Granada, Spain; [k]Brain, Mind and Behaviour Research Center (CIMCYC), University of Granada, Granada, Spain; [l]Instituto de Investigación Biosanitaria de Granada, IBS, Granada, Spain

ABSTRACT

Purpose: This study presents a preliminary multi-stakeholder needs assessment to inform the subsequent co-design of a digital telerehabilitation tool aimed at supporting organisation, planning, and prospective memory in children and adolescents with ADHD. The primary outcome was to identify functional challenges, environmental determinants, and user-reported priorities across stakeholder groups. The secondary outcome was to translate these needs into preliminary design requirements and guiding principles for future co-design phases.

Methods: A qualitative study was conducted using focus groups guided by the CFIR framework, involving 49 participants: children and adolescents with ADHD, caregivers, occupational therapists, and educational professionals. Data were analysed through deductive thematic coding and inductive grounded-theory synthesis. Matrix mappings and Sankey diagrams were used to visualise convergence and divergence across stakeholder perspectives.

Results: Three cross-cutting priority areas emerged: (1) emotional self-regulation, (2) autonomy in daily routines through customisable reminders and supports, and (3) psychoeducation for families and educators. Children emphasised reminders, prospective-memory support, privacy control, and motivational feedback. Educators highlighted the need for training and classroom-based co-regulation strategies. Occupational therapists emphasised ecological, participation-oriented interventions. Caregivers reported emotional burden, concerns about independence, and a need for accessible psychoeducational resources.

Conclusions: This needs assessment provides a structured, replicable foundation for the co-design of a digital telerehabilitation tool for ADHD, integrating emotional, functional, and educational dimensions. Future research should evaluate the feasibility, usability, and the impact of the resulting prototype on autonomy, participation, and family dynamics.

➤ IMPLICATIONS FOR REHABILITATION

- The study proposes a reproducible, collaborative model for designing digital telerehabilitation that address three key priorities: emotional self-regulation, independence in daily routines, and psychoeducation for families and educators.
- Children and adolescents emphasised the need for customisable reminders, privacy controls, and motivational feedback.

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Attention deficit disorder with hyperactivity; self-regulation; telerehabilitation; user-centred design; executive function

CONTACT Dulce Romero-Ayuso dulceromero@ugr.es Facultad de Ciencias de la Salud, Universidad de Granada, Avda. de la Ilustración n° 60, Granada, 18600, Spain

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- The results highlight the systemic impact of ADHD and the importance of family-centred support in digital health solutions for their rehabilitation.
- Professionals underscored the importance of ecological interventions and the development of self-regulation skills.

Introduction

Attention-deficit hyperactivity disorder (ADHD) is one of the most prevalent neurodevelopmental disorders in childhood and adolescence, affecting ~5–7% of the school population [1]. This disorder is characterised by persistent patterns of inattention, hyperactivity, and impulsivity that significantly interfere with academic, social, and family functioning. Executive dysfunction in ADHD has been widely documented as a contributing factor to difficulty in behavioural and emotional self-regulation, as well as in the execution of complex tasks that require sequential planning and inhibitory control [2]. People with ADHD often experience difficulties in perceiving time, initiating tasks, and maintaining focus on future goals, which can lead to feelings of overwhelm and low self-esteem [3].

Although pharmacological treatments, particularly stimulants, are currently the first therapeutic choice, their impact on executive functions (EF) has proven to be limited [4,5]. In addition, many non-pharmacological tools used in assessment and intervention lack ecological validity and have limited applicability in everyday contexts. A large part of neurorehabilitation programs has focused on decontextualised cognitive functions, without adequately considering the user's functional environment or daily experience.

In response to these limitations, the concept of functional cognition has gained relevance in both applied neuropsychology and occupational therapy (OT). This approach integrates cognitive functions with the effective execution of meaningful tasks in real environments and allows assessing how a person applies skills such as prospective memory (PM) or organisation in their daily lives. PM, understood as the ability to remember future intentions at the right time [6], is crucial in activities that require temporal coordination or reactions to specific events, such as turning in homework, taking medication, or going to an appointment. Their deterioration can lead to disorganised functioning and over-reliance on adult figures.

Studies have shown the effectiveness of various interventions and support systems in helping children and adolescents with ADHD develop better organisational and time management skills [7,8]. Implementing specific strategies, such as breaking tasks down into smaller components, using visual supports, and establishing clear prioritisation systems, is especially beneficial for this population [9]. In addition, multimodal approaches that combine various interventions, such as parent training programs and social skills training, have shown positive results in improving both emotional symptoms and daily time management skills [8]. These approaches, when combined with appropriate goal-setting techniques and consistent routines, contribute to creating a favourable environment for successful task completion and better self-regulation [4,5]. A meta-analysis on organisational skills training (OST) for children and adolescents with ADHD revealed moderate to large improvements in organisational skills, as assessed by teachers and parents, as well as moderate improvements in inattentive symptoms and academic performance [4]. Taken together, organisational skills interventions appear to be an effective approach to addressing ADHD-related impairments in academic and daily functioning.

OT plays a key role in addressing these challenges by focusing on improving children's functional participation and independence in meaningful contexts, helping them to develop an appropriate sense of competence. However, access to them may be limited by economic, logistical, or geographical barriers, especially in rural areas or in low-income families [10]. Faced with this situation, the potential of emerging technologies such as virtual reality (VR) and personalised tele-rehabilitation platforms has begun to be explored. These tools allow EF to be evaluated and trained in controlled environments with high representation of real life, favouring transfer to everyday contexts [11,12], and offer an accessible alternative to traditional limitations on access to specialised services. In this context, digital technologies have emerged as a promising alternative to complement traditional treatments, offering opportunities to personalise, scale, contextualise, and monitor the results of interventions in real time [13,14]. Digital

interventions can be developed through mobile phones, tablets, or web platforms providing interventions that are more accessible, better integrated into multiple settings (e.g., home, school, and health services), and enable individuals (and families) to take care of themselves [15]. Digital therapies have the potential to be implemented anytime, anywhere, which could ease the burden on caregivers when organising interventions for their children with ADHD. In turn, this can improve patient adherence to the intervention and generate long-lasting therapeutic effects [15]. Likewise, the use of digital games is an effective strategy to increase adherence to interventions in the child population. By offering an interactive and engaging environment, these resources can enhance children's motivation to continue treatment, while allowing for appropriate feedback during their development [16]. On the other hand, these interventions can be easily adapted to the specific needs of each user, promoting participation, motivation, and personalisation based on different levels of development and individual preferences, depending on the areas in which they present deficits [17,18].

The rise of therapeutic and gamified digital interventions for ADHD has shown that technological platforms can modulate attentional processes and certain executive dimensions, although evidence specifically oriented to organisation and planning in child and adolescent populations remains incipient and fragmented [13]. For example, some gamified platforms are highly effective in improving cognitive and behavioural skills, encouraging participation in structured activities in playful environments that reinforce positive behaviours [19,20]. However, these studies have relevant methodological limitations, such as small sample sizes, scarce information on the implementation context, and a limited description of the specific functionalities of digital tools. This underscores the need to develop more robust research that not only evaluates the efficacy of these interventions but also actively integrates end-user perspectives into their design and validation. In this sense, co-creation and co-design, as participatory approaches in the development of health technologies, make it possible to involve children and adolescents with ADHD, their families, occupational therapists, educators, and technology developers in a collaborative process that favours the relevance, usability, and acceptance of digital solutions [14,21,22].

Within the ADHD literature, difficulties in organisation and planning are understood as core components of executive functioning and functional cognition [5,23]. These skills depend on processes such as working memory, prospective memory, inhibitory control, and self-regulation, which together support the ability to initiate, sequence, monitor, and complete daily activities [6]. In children and adolescents with ADHD, impairments in these domains often translate into challenges in managing routines, completing school tasks, and maintaining autonomy in daily life [24,25]. For this reason, the present study focuses specifically on organisation and planning skills as the primary functional targets for a future digital telerehabilitation tool, while acknowledging the broader executive processes that underpin them.

Given this convergence of clinical needs, technological opportunities, and user-centered design principles, this paper presents a preliminary needs assessment to inform the co-design of a digital telerehabilitation tool for children and adolescents with ADHD, with a particular focus on organisation and planning skills. Specifically, the study aims to: (a) identify and prioritise functional needs and daily challenges as perceived by key stakeholders; (b) describe the participatory qualitative methodology used to elicit these perspectives; and (c) translate stakeholder input into user-informed design requirements and guiding principles. By systematically integrating the perspectives of children and adolescents, families, educators, and clinicians, this study provides a robust empirical foundation for the development of meaningful, feasible, and user-centered digital interventions for ADHD.

Methodology

Study design

This qualitative study was conducted within a user-centered design framework and used focus groups to engage four key stakeholder groups: children and adolescents with ADHD, their primary caregivers, occupational therapists, and educational professionals with experience and direct contact with these children and adolescents. The CFIR was used to guide the development of the discussion guide, as it conceptualises barriers and facilitators as implementation-relevant determinants, providing a structured framework to prospectively explore contextual, organisational, and individual factors that may influence the future

adoption, implementation, and sustainability of an intervention [26]. This structure provided a theoretically grounded rationale for selecting and organising the domains explored in the focus group questions. CFIR served as the determinant-based structure for the discussion guide, organising questions around the domains of Intervention Characteristics (adaptability, complexity, perceived advantage), Individual Needs and Resources, Outer Setting, and Process. This structure ensured that all stakeholder groups were systematically asked about barriers, facilitators, contextual factors, and implementation needs relevant to the future implementation of the digital intervention. A detailed mapping of the discussion questions to CFIR domains and thematic blocks is now provided in Supplementary Table S1.

Ethical approval was obtained from the Ethics Committee of the Research Centre of Granada (CEI-GRANADA), and written informed consent was provided by all participants. The study complied with the Declaration of Helsinki and Spanish Law 14/2007 on Biomedical Research.

Focus groups and preparatory meetings were held at the Faculty of Health Sciences of Granada, Spain, in accessible ground-floor laboratories designed to offer a comfortable and familiar environment. A total of 49 participants were included in eight focus groups conducted in May and June 2024: two with children and adolescents with ADHD ($n = 8$), two with caregivers ($n = 11$), two with occupational therapists ($n = 20$), and two with educational professionals ($n = 10$, whom 8 were teachers and 2 were school counsellors). Although formal data saturation was not evaluated, the number of groups aligns with established guidelines, indicating that 80–90% of key themes typically emerge within 2–6 focus groups.

Participant recruitment

Purposive sampling was employed to ensure representation from all stakeholder categories. Children and adolescents and their primary caregivers were recruited through the Specialised Educational Guidance Team for Severe Behavioural Disorders of Granada (Ministry of Educational Development and Vocational Training, Regional Government of Andalusia), which distributed invitations through school guidance staff. Additional recruitment was supported by neuropediatricians from the Virgen de las Nieves Hospital, paediatricians from the Zaidín Sur health centre, and families involved in a previous AMPA-linked project.

Educational professionals were also recruited through the Educational Administration, specifically *via* the same Specialised Educational Guidance Team. Occupational therapists were contacted through the Spanish Council of Occupational Therapy Colleges, the Spanish Scientific Society of Occupational Therapy for Children and Adolescents (TOPES), and institutional mailing lists. Some professionals were contacted directly based on known experience with neurodevelopmental disorder (NDD) populations.

Recruitment followed a three-step process: (1) Initial contact *via* email or phone, providing study information and confirming voluntary participation. (2) Up to three in-person meetings at the university to clarify objectives, resolve doubts, and obtain written informed consent (and assent in the case of children and adolescents). (3) Assignment to one of eight scheduled focus groups, ensuring balanced composition.

A total of 72 formal invitations were extended across the four participant categories. Sessions were strategically scheduled on different dates and at varying times to accommodate participants' personal and professional commitments. Each stakeholder group (children and adolescents, caregivers, occupational therapists, and educational professionals) were invited to participate in two specific sessions, allowing for greater flexibility and facilitating participation. This staggered and targeted scheduling approach enhanced attendance and ensured a balanced representation across all stakeholder groups.

Although not all invitees attended, this multi-date strategy optimised participation and group diversity. The final sample comprised 49 participants: 8 children and adolescents, 11 caregivers, 20 occupational therapists, and 10 educational professionals (8 teachers and 2 school counsellors). A flowchart detailing recruitment and inclusion is provided in Figure 1.

Inclusion and exclusion criteria

Children and adolescents were eligible to participate if they were between 9 and 17 years old, fluent in Spanish (spoken and written), had an educational or clinic diagnosis ADHD, an intelligence quotient (IQ) above 69, and scored equal to or above 65 on both the Cognitive Regulation Index and the Behavioural

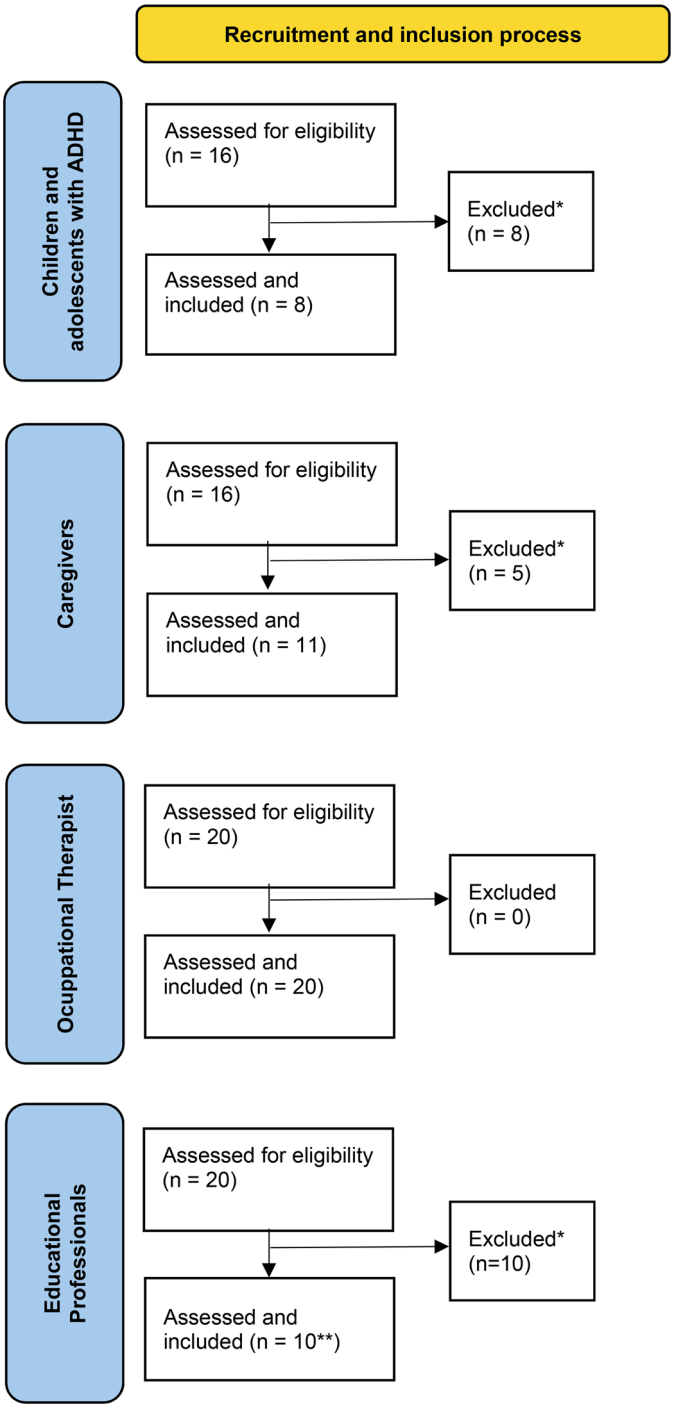


Figure 1. Recruitment and inclusion process.
Note: All caregivers in the sample were the primary caregivers of the participating children and adolescents; thus, the study included complete child–caregiver dyads
*He/She announced on the focus groups day that he/she would not be able to come.
**8 teachers, 2 school counsellors.

Regulation Index of the BRIEF-2, as reported by parents or teachers. Additionally, they were required to provide written informed assent, along with guardian consent.

The selection of the 9–17 age range was grounded in developmental and functional considerations. This interval was selected because it encompasses a developmental period in which executive functions (e.g., planning, working memory, inhibitory control, and self-regulation) undergo substantial maturation and show considerable interindividual variability [27,28]. Including participants across this span allows

the identification of functional challenges and everyday demands characteristic of both late childhood and adolescence [29,30]. From approximately age 9 onward, students encounter increased academic complexity, heightened organisational expectations, and a progressive need for autonomous task management; these demands intensify during adolescence and directly influence daily functioning in individuals with ADHD [31]. Therefore, the 9–17 range aligns with the evolving contextual challenges that this population faces in school, home, and community settings [30]. In addition, children aged 9 years and older generally demonstrate sufficient digital competence to participate meaningfully in user-centered design activities and needs-assessment procedures, while adolescents up to 17 years typically report extensive exposure to digital technologies, enabling them to provide detailed and relevant feedback regarding usability, acceptability, and functional requirements—key components in the early stages of tele-rehabilitation development. Finally, the selected age span captures a developmental window in which needs related to time management, planning, task initiation, emotion regulation, and autonomy change substantially; including this period facilitates the identification of both age-dependent and age-invariant needs to inform subsequent phases of tool refinement and evaluation [27,28].

Exclusion criteria for this population were the presence of sensory or motor limitations that preclude the use of electronic devices (e.g., tablets, computers, or smartphones), documented photosensitive epilepsy, or uncontrolled aggressive behaviour. Importantly, children presenting visual hypersensitivity or visual-seeking behaviours were not excluded based on these sensory characteristics alone, as such profiles are common in ADHD [32,33] and directly relevant to the co-design and preliminary needs assessment of a visually delivered telerehabilitation tool. Uncontrolled aggressive behaviour was assessed through caregiver reports of incidents occurring within the previous three months (e.g., episodes resulting in injury or property damage, emergency intervention, or school suspension) and, when applicable, clinician input. Exclusion due to photosensitive epilepsy was determined based on medical records or caregiver/participant reports of prior seizures triggered by visual stimuli, clinical advice to avoid screen exposure, or a positive response to a pre-session screening item indicating seizures, migraine with visual aura, or discomfort induced by flicker or visual patterns. For sensory or motor limitations, verification relied primarily on clinical documentation available at referral, and this information was subsequently confirmed with caregivers during the enrolment visit before consent.

Caregivers were eligible if they were legal guardians, fluent in Spanish (spoken and written), co-resided with the child/adolescent at least 50% of the time in cases of shared custody, expressed a clear willingness to participate, and signed a written informed consent form. All caregivers who participated in the study were the primary caregivers of the children and adolescents included in the child/adolescent focus groups. No caregivers whose children did not participate were recruited. Each child/adolescent had a corresponding caregiver who attended the caregiver focus group, ensuring complete dyadic representation across stakeholder sessions.

Professionals (occupational therapists and professional educators) were included if they had a minimum of five years of experience working with children or/and adolescents diagnosed with ADHD, were currently employed in educational or clinical settings, were fluent in Spanish (spoken and written and provided written informed consent).

Discussion guide

The development of the focus group questions followed established qualitative research guidelines ensuring an open, flexible script suitable for diverse participants. We first identified cross-cutting themes relevant across stakeholder groups primarily experiences with technology and digital therapeutic tools [34] to ensure conceptual coherence and facilitate comparison across stakeholder perspectives. Guided by these initial themes, the research team developed a core set of questions addressing:

- Daily life challenges and support needs
- Desired design and usability features
- Preferences for content organisation (levels, themes, activity types)
- Feedback and progress-tracking expectations

- Motivational systems
- Privacy and data protection expectations

The structure and wording of the questions were refined through five iterative team meetings held between January and June 2024. To standardise phrasing and ensure systematic alignment with implementation constructs, we incorporated the CFIR Interview Guide Tool (https://cfirguide.org/guide/app/index.html), which supported script refinement and ensured comprehensive coverage of the following domains: Intervention Characteristics, Individual Needs and Resources, Outer Setting, Inner Setting, and Process.

Building on this framework, group-specific questions were developed for children/adolescents, caregivers, teachers, and therapists. Children and adolescents, as well as caregivers, were asked prompts such as: "What would you like the app to include?", "How could it help you in your daily life?", and "Would seeing your progress after each session be useful?". Professionals were additionally asked about the potential integration in practice, priority content areas, appropriate automated features (e.g., reminders), and strategies for delivering meaningful feedback. They were also invited to identify outcome domains relevant to future clinical trials, including clinical, behavioural, usability, autonomy, and daily functioning metrics.

Given the exploratory nature of this study, caregiver burden was not included as an independent theme. However, issues related to time demands, feasibility, and routine organisation emerged naturally within several CFIR domains and were explored through moderator-led probes as participants raised them. Moderators used targeted probes to elicit practical strategies to minimise potential caregiving burden, including preferred session scheduling and frequency, coordination with school/family routines, division of responsibilities, reminders/automation, and acceptable time-on-task and reporting requirements. A detailed mapping of questions to CFIR domains and thematic blocks is provided in Supplementary Table S1.

Each focus group lasted ~90–120 min. All sessions were audio-recorded and transcribed verbatim. Facilitators also took observational notes to capture non-verbal cues, group dynamics, and contextual aspects relevant to interpreting the qualitative data (Figure 2).



Figure 2. Roadmap of the preliminary needs assessment with children and adolescents with ADHD and key stakeholders.The purple box emphasise the key aspects necessary to understand the study and the central research questions addressed. The green boxes illustrate the chronological sequence of events throughout the study, including recruitment, meetings, and other relevant activities.

Data analysis

A comprehensive two-phase analytic strategy was implemented to ensure a robust and nuanced understanding of the qualitative data. In the first phase, a deductive thematic analysis was conducted, guided by the Consolidated Framework for Implementation Research (CFIR). This framework provided a structured lens for the initial coding process, allowing the research team to categorise data according to established domains and constructs relevant to implementation science [26,34].

In the second phase, an inductive analytic process informed by grounded theory principles was employed to synthesise individual codes into higher-order themes, allowing novel and context-specific patterns to emerge beyond the predefined CFIR categories.

Following this synthesis, the resulting higher-order themes were subsequently situated within the WHO International Classification of Functioning, Disability and Health (ICF/ICF-CY), with particular emphasis on Environmental Factors (e1 products and technology; e2 natural environment; e3 support and relationships; e4 attitudes; e5 services, systems and policies), to foreground environmental determinants of participation [35]. In parallel, these higher-order themes were cross-walked to the family of Participation-Related Constructs (fPRC)—including attendance, involvement, activity competence, sense of self, preferences, and environment—to enhance interpretive depth regarding participation processes and outcomes [35].

To enhance the credibility and trustworthiness of the findings, the coding process was carried out independently by three researchers with expertise in qualitative methods. Regular meetings were held to discuss discrepancies, refine coding categories, and reach consensus, thereby minimising individual bias and promoting inter-coder reliability. All qualitative data were systematically managed and organised using ATLAS.ti software (version 9.1), which facilitated the coding process, supported the retrieval of coded segments, and enabled the transparent documentation of analytic decisions throughout the study. In addition to coder triangulation, several integrative procedures were implemented to synthesise perspectives across stakeholder groups. Analyst triangulation was carried out among researchers from rehabilitation medicine, occupational therapy, neuropsychology, and education, ensuring multidisciplinary interpretation of the data. To systematically compare viewpoints, we developed matrix mappings of themes by stakeholder group (youth, caregivers, teachers, and therapists), which enabled the identification of convergent, divergent, and complementary needs. A series of consensus sessions was then conducted to consolidate these themes into higher-order categories aligned with the CFIR domains. Subsequently, integration sessions were held to merge convergent priorities into shared design principles (e.g., usability, flexibility, personalisation, ecological validity). Finally, a model-building phase synthesised these insights into a preliminary conceptual model of the digital intervention, ensuring that the needs assessment process resulted in a coherent and theoretically grounded framework. The analysis followed a hybrid approach: CFIR domains were used to identify implementation determinants (barriers and facilitators), while the fPRC and ICF frameworks categorised the functional needs and participation factors related to organisation and planning. Following the synthesis of individual codes into higher-order themes, a brief analytic memo was used to guide the interpretive mapping of these themes to ICF Environmental Factors (e1–e5) and to fPRC constructs. Empirical literature on environmental determinants of participation in ADHD (e.g., PEM-CY–based studies across home, school, and community) was used as a sensitising corpus to support the interpretation and labelling of barriers and facilitators (e.g., cognitive demands, sensory qualities, teacher/peer support, organisational and policy-related features) [36]. A cross-walk of higher-order themes to ICF domains (including Environmental Factors e1–e5) and fPRC constructs is provided in Supplementary Figure S2.

Positionality

The research team comprises ten members from two Spanish public universities with experience in the field of neurodevelopmental disorders. Six members identify as women and four as men. The women include a physiatrist (MR-G); three occupational therapists (CVR, MMT, MCRM); a teacher and educational psychologist (EMT); and an occupational therapist and neuropsychologist (DRA). The men include a teacher and pedagogue (JAMM); an occupational therapist (ATG); a primary care physician specialised in

preventive medicine and public health (JMTJ); and a computer scientist (MGM). All members have backgrounds in research and practice across clinical and/or educational settings.

The team acknowledges that its training and practice contexts (e.g., occupational therapy, medicine—rehabilitation, primary care, preventive and public health, psychology/neuropsychology, education, and computer science) shape interpretive lenses and may influence how questions are framed, how experiences are interpreted, and which needs are prioritised. Working within Spanish public health and education systems may heighten attention to organisational feasibility, integration with public services, equity of access, and acceptability among families and professionals. Clinical and educational profiles may also orient discussions towards functional and participation outcomes, as well as usability and real-world applicability.

To mitigate potential bias arising from this positionality, the team implemented explicit safeguards in the design and facilitation of focus groups: (a) a CFIR-informed discussion guide with open, comparable questions across stakeholder groups; (b) reflexive moderation with predefined probes to surface divergent perspectives, caregiving burden, and burden-reducing strategies; (c) analyst triangulation across disciplines; and (d) internal auditability of analytic decisions *via* consensus meetings, memos, and coding matrices. Positionality is approached as both a resource and a responsibility, making the team's lenses transparent so readers can assess the scope and boundaries of the interpretations.

Results

The coding analysis carried out on the qualitative corpus allowed us to identify the distribution of relevant topics, from their experiences and needs to the conceptualisation of a digital tool for children and adolescents with ADHD in relation to the main actors involved: children and adolescents, teachers, occupational therapists, and mothers/fathers/caregivers. Visualisation using a Sankey diagram made it easier to observe how the emerging codes are linked to each group, highlighting multiple relationships, overlaps, and specific focus of attention (Figure 3). These descriptive patterns were later synthesised into higher-order themes and interpreted within the WHO ICF/ICF-CY—particularly Environmental Factors (e1–e5)—and the Family of Participation-Related Constructs (fPRC), which guided the identification of participation-related needs across stakeholder groups. These descriptive patterns also reflected the CFIR-informed structure of the discussion guide, with codes aligning with domains such as Intervention Characteristics, Outer Setting, Characteristics of Individuals, and Process, as detailed in Supplementary Figure S1.

Children and adolescents

This group presented a high density of connections, highlighting the codes linked to EF, emotional regulation, and behavioural aspects. Among the most frequent topics about their needs were: prospective memory, time management, use of reminders, and personalised alarms to carry out meaningful activities (ADLs, school, and leisure activities). Within this topic, a child indicated: "*Yes, a place that helps you do things and gives you advice on how to do them.*" *You can add photos as reminders, and if you forget, an audio alert plays to tell you that the time has passed. And if you forget, and alarm goes off.*" *It would be necessary to include session reminders*." Another indicated that it would be useful: "*A mini-game section where you can choose games to improve memory*." Regarding the need for help to remember the tasks they must perform, one child commented: " *Yes, writing down your routine, for example what you have to do at three o'clock or at six o'clock, would be really useful. And if you forget something you scheduled for a certain time, can the app remind you (…). I mean, you press a button and a timer starts, depending on how much time you have for that activity, it gives you the time and reminds you if you forget*." The need for daily reminders was related to the use of customisable alarms: " *For example, it could say: at three o'clock, make your bed. If you forget, an alarm goes off. Then at four o'clock, it reminds you to do your homework*", "*or that it reminds you that you have to go to sleep or other things.*" In the group of children and adolescents, concern about carrying out activities was also shared and was related to prospective memory and time management skills: "*It would be useful to include time management activities*." Another commented: "*For example, I have difficulty studying and retaining what I've just read.*", "*I would like it to help me have more responsibility. Reinforce the things that I am not good at. Be smarter. To help me remember things*."



Figure 3. Sankey diagram illustrating code concurrence across the different discussion groups.

From a CFIR perspective, these child-reported needs align with Characteristics of Individuals (knowledge, beliefs, self-efficacy) and Intervention Characteristics (design quality, adaptability), shaping how children envisioned the tool's usefulness.

Related to this aspect, another adolescent indicated: "*If I have a reminder that I don't want my parents to see, I should be able to hide it or turn it on and off.*" suggesting that the intervention should safeguard the autonomy, sense of competence, and privacy of the participants.

The second theme that emerged from the children and adolescents was the need to develop a section to improve their skills in planning, organising, and supervising action. One of the participating adolescents indicated that the intervention should: "Include exercises that *address organisation and planning*." Another commented in relation to the activities: "*putting myself in a daily routine, of tasks, advice, that is, help so that they give you tricks, minigames (…)*". And another commented: "*Yes, yes, Setting up a daily routine with tasks, advice, tips, and mini-games,…*" In relation to the supervision of the action and feedback on the performance of activities, they indicated: "*I would like you to say very well, you are improving your statistics a lot*".

The third suggested topic was psychoeducation, so some adolescents indicated that it would be useful: "*Videos on how to do things, for example, how to organise your table, desk…, demonstration videos*".

The fourth theme that emerged from the group of children and adolescents was related to self-regulation, awareness, and self-esteem. One child indicated that he would like the intervention to help him "*Know when you need a break. To get to know each other and to know what kind of activities are good for me to rest (close my eyes for a few seconds, move)*." Another teenager commented, "*Some kind of alert or guidance to encourage movement breaks could be included*." Another adolescent commented that the use of alarms or time management is sometimes related to emotional and cognitive

regulation: "*If they give you a set amount of time, sometimes you focus more on the time left than on doing the task properly*". Another participant indicated that he would like *"at the beginning I would like him to ask me something personal: How did you spend your time in class today?, ¿How were you with your friends?, is it important for self-esteem…"*.

These needs mapped primarily onto ICF Environmental Factors related to support and relationships (e3) and products and technology (e1), as well as fPRC constructs such as involvement, activity competence, and preferences.

Educational professionals

The connections with this group revolved around aspects of institutional support, accompaniment, and training. Two major themes stand out: (1) Emotional regulation and training in strategies for Self-Regulation; (2) Need for teacher training in specific methodologies due to attentional difficulties, constant supervision of tasks, memory, and planning skills. This suggests a double need: on the one hand, for specific strategies that can be implemented in the classroom, and on the other, for training and accompaniment for teachers to understand and respond adequately to the needs of students with self-regulation and planning difficulties. One teacher indicated: "*It would be useful to implement routines at school, as students can become easily distracted, and to look for alternatives such as rest, exercise, or mindfulness when they reach a saturation point*." Regarding the difficulties in attending, another teacher said: "*They don't listen, they get bored. I think they are bored too*." "*There may be considerable comorbidity with reading and writing difficulties, as some students struggle to learn to read properly*." "*It would have to be something that would also serve for them to be successful in school. So perhaps there would be a study technique, for example, that could be specific for this type of student. Something that would help them, a methodology*." Another educator also expressed: "(…)*I would love cutting-edge articles on ADHD… I do too. Things like this… And also podcasts… I'd love to. (…). On scientific evidence, real testimonies.*"

Another participant indicated the need to address planning skills and their relationship in emotional regulation: " *Planning, which is the focus of the study, and organisation, and, perhaps, a lack of emotional control, which is a big part of the problem, isn't it? That is, emotional instability that affects people. Emotions that don't know… Many children experience emotions that they do not know how to identify or recognise. And, from there, manage them*." In this area, an educator highlighted the importance of involving the family with a specific section on psychoeducation: *" For families, a section with interesting pages and information from leading paediatricians. And, the emotional part, emotional management with training, right? For example, emotional management in the face of adversity that can occur in an emergency. (…) That is, where do I turn when this happens with the child? It would be interesting to have a format, a medium, a guide for internal communication of those who are families, guardians or parents among them*."

These data show that educational professionals not only act as direct observers of students' difficulties but also require specific support and strategies to be able to intervene appropriately. The presence of terms such as "constant supervision" or "orientations" suggests the need for more systematic vocational training and institutional support. These patterns correspond to CFIR domains related to Outer Setting (student needs and resources) and Intervention Characteristics (complexity, relative advantage), highlighting the institutional dimension of implementation. From an ICF perspective, these concerns align with Environmental Factors e3 (support and relationships), e4 (attitudes), and e5 (services, systems, and policies), highlighting how school-level structures shape participation opportunities. Within the fPRC, these issues relate to environmental supports influencing attendance and involvement.

Occupational therapists

The analysis showed numerous codes in this group, reflecting their integral role. The most prominent connections were for self-regulation, activities of daily living, psychoeducation, and child/adolescent motivation. These relationships reflect their participation in the design and implementation of functional and adaptive interventions, as well as their contribution to the training of specific skills for the child's autonomy. There is also an important intersection with psychoeducation and self-regulation, which underlines their contribution to a comprehensive approach to support. These observations map onto

CFIR domains concerning Characteristics of Individuals (professional knowledge and beliefs) and Intervention Characteristics (adaptability, design quality), reflecting therapists' dual role as implementers and co-design contributors.

Regarding self-regulation, several occupational therapists indicated the importance of considering it in a transversal way: *"That is, they do not understand that they sometimes need to get up and regulate themselves. Often, no other regulatory strategies are offered, and the response is simply to tell them to 'sit down, sit down'. I think this needs to be addressed through a more multidisciplinary approach involving the teacher, the therapist, and the child, in order to better understand self-regulation, which is ultimately very important."* Another therapist noted, *"It's all part of self-regulation. Instead of having a single category for it, self-regulation [strategies] should be integrated within each specific section, such as emotional, sensory, or motor coordination."* Another pointed out: "*So by doing it in reverse in each subsection, keep the self-regulation section*."

In relation to the activities of daily life, they indicated that it is important: "*to recreate these situations of daily life virtually and to be able to practice and train them at home, right?*". This topic was related to planning skills in schoolwork: "*One of the workhorses is, how do you organise the work that has been given to you at school, at home? In other words, at school they give you homework and then you take home the homework to do and now everything at home you must organise, right?*". Other therapists related it to motivation: "*They tell me (…) that doesn't stimulate me, I've been working with an agenda for a year and now, well, it doesn't stimulate me. I need something that really stimulates me, well, so that they don't have to repeat me, take a shower, the parents take a shower, brush your teeth, right? (…) go to the bathroom, such, that whole daily routine*." Another therapist pointed out: "*To achieve the good intervention, a high level of motivation and interest is needed, and in the end, adolescents have more motivation and interest in the screen. (…) I, for example, try to use it as little as possible, but there is perhaps in some cases, especially in food, that in the end the final reward of the last five minutes is the tablet. Because in the end, it's the reward that they want, that is, the highest motivation is that. So, really, I think it would be ideal to make good use of the screen, but to do the intervention through technology, which in the end is what they want and is part of their daily lives*."

As in the case of educational professionals, occupational therapists considered it necessary to include psychoeducation for parents: "*something that can perhaps be called, psychoeducation, some guidelines can be given to parents, or that in this user profile of parents, well, there is that type of guide or even example videos, (…), and also that there is audiovisual material with feedback (…)". Because maybe the parents themselves can have a community because we know that they need psychoeducation and support, and on many occasions, it does not reach everyone, very few*. "These results are reflected graphically and more extensively in Figure 4. These observations correspond to ICF domains involving products and technology (e1) and natural environment demands (e2), and to fPRC constructs such as activity competence and environmental fit.

Parents

The group of parents showed especially relevant associations with emotional issues and family burden. Among the most frequent topics are: (1) Parental anxiety and refusal of children/adolescents to go to school; (2) Sense of competence; (3) Overload of family members with homework; (4) Lack of communication; and (5) Lack of understanding about the circumstances of the child/adolescent and need for psychoeducation. This indicates that the family, in addition to acting as a support network, experiences a strong emotional and functional impact in the face of the child/adolescent's demands, which highlights the importance of including caregivers in the therapeutic and educational process in an active and collaborative way. The family, in addition to observing the difficulties of the child/adolescent, experiences high levels of stress and burden, which can negatively impact the family dynamics. The perception of parental self-efficacy and the need for concrete tools to support the child are key elements that emerge from this analysis. These family-level concerns align with CFIR Outer Setting constructs (patient needs and resources) and Characteristics of Individuals (self-efficacy), underscoring the centrality of caregiver perspectives in implementation feasibility.

Regarding the anxiety and concern of parents about the achievement of autonomy of their children, some parents commented: *"In the future you will have to keep struggling, right?", "My God, I am trembling*

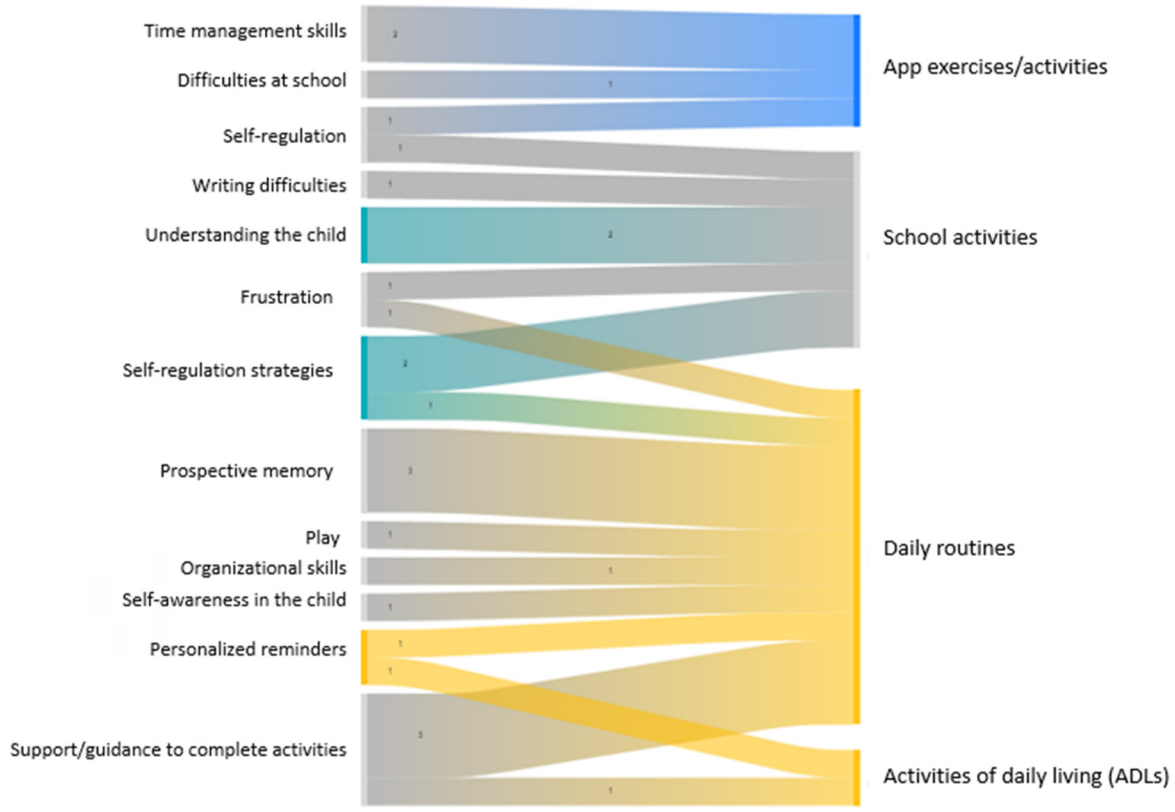


Figure 4. Sankey diagram illustrating the thematic concurrence within daily activities.

from ESO (Compulsory Secondary Education)". Another parent commented, *"As parents, right? We have not been so much concerned with the quality of what he has done, but with the establishment of the conduct. That is, at the beginning, well, that the bed was made, even if it was not done well."* This issue was also related to the lack of awareness that children have of: *"the difficulty I find with my children is that they create that responsibility for them to be aware, no matter how much I repeat every day." "That they are aware."*

Likewise, some relatives commented on the difficulties that the children/adolescents to have friends at school and even suffer situations of bullying: "*It can be affected because they are being bullied at school, they stop attending".* Another parent indicated the effort that his son makes to self-regulate at school and the lack of understanding and resources in it: "*But he tries not to have problems at school", "Because at school, when a child is telling you five hundred thousand times and he doesn't know and he is not doing his homework, it may be because [the teachers] cannot always keep an eye on him they cannot be aware of him."*

In relation to the need for psychoeducation, the parents indicated: "*because you are asking questions that come to you, that you don't know, let's see you don't have anyone to ask every day, or you Google it, but many times maybe someone who has already gone through that or what is happening will answer you, hey, well, this is working for me, or I have gone to the neurologist and he has told me this, and, well, that would be fine".*

These family-level challenges reflect ICF Environmental Factors e3 (support and relationships) and e4 (attitudes), and relate to fPRC constructs such as sense of self, preferences, and involvement, which shape participation across home and school contexts.

Across stakeholder groups, conceptually related codes and descriptive themes converged around a set of cross-cutting domains. Through iterative axial coding, these patterns were synthesised into higher-order themes that integrated perspectives across groups and contexts. For example, codes referring to emotional regulation difficulties, anxiety, and the need for constant supervision across home and school settings were clustered into the higher-order theme "Everyday challenges and support needs". Similarly, codes related to interface simplicity, reminders, customisable alarms, and accessibility features were synthesised into the theme "Design and usability features". This analytic process was applied iteratively across the full set of codes. These cross-group patterns also reflected CFIR domains identified during

initial coding, particularly Outer Setting (needs and resources), Intervention Characteristics (design features, complexity), and Process (reflecting and evaluating), which structured the emergence of higher-order themes.

The resulting higher-order themes and their illustrative examples were subsequently interpreted within the WHO ICF/ICF-CY and the family of Participation-Related Constructs (fPRC) frameworks, as detailed in Supplementary Figure S2.

Overall, the results show that the design of the tool and intervention should contemplate: 1) Emotional components, including the emotional regulation of the child and family, self-esteem and sense of competence in the daily life of the child/adolescent; 2) Cognitive training: attentional, in EF and prospective memory (planning, organisation, reminders); 3) An educational dimension (psychoeducation, understanding of the child/adolescent) aimed at both parents and educators. In addition, it must be customisable and adaptable to different user profiles, integrating the perspectives of all the actors involved in the rehabilitation process.

From identified priorities to design features

Across stakeholder groups, the three cross-cutting priorities—emotional self-regulation, independence in daily routines, and psychoeducation—were consistently associated with concrete expectations about how a digital tool should function in everyday contexts. These expectations emerged spontaneously in participants' narratives and reflected patterns aligned with CFIR domains and fPRC constructs.

For emotional self-regulation, children, caregivers, teachers, and therapists described the need for features that help identify, express, and modulate emotions in real time. Participants proposed emotion-labelling interfaces, short, guided strategies (e.g., breathing, movement breaks), and prompts that support self-awareness. These expectations corresponded to CFIR Intervention Characteristics (adaptability, perceived advantage) and Individual Needs and Resources and mapped onto fPRC constructs related to participation—particularly involvement and sense of self—as children emphasised the importance of feeling understood, supported, and autonomous.

For independence in daily routines, participants described highly specific supports for planning, organisation, and prospective memory. Children repeatedly mentioned customisable alarms, visual planners, and step-by-step task sequences to scaffold homework, self-care, and daily responsibilities. Parents and therapists reinforced the need for tools that enhance activity competence and reduce family burden, reflecting CFIR Outer Setting determinants (family needs and resources) and Process elements related to monitoring and feedback. These expectations aligned with fPRC constructs of activity competence, preferences, and attendance, as children expressed a desire for tools that were motivating, personalised, and responsive to their routines.

For psychoeducation, all groups—especially parents and educators—highlighted the need for accessible, evidence-based information to better understand ADHD, executive functioning, emotional regulation, and effective support strategies. Participants requested short videos, demonstrations, curated resources, and guidance tailored to different user profiles. These expectations reflected CFIR Intervention Characteristics (complexity, design quality) and Outer Setting needs for coordinated support across home and school.

Finally, from an fPRC perspective, these needs related to the environment dimension—particularly the desire for supports that were acceptable, accessible, accommodating, affordable, and available for families and professionals

Discussion

This study delimits, based on the experience of children and adolescents with ADHD, their families, educational professionals, and occupational therapists, the contents and principles of design of a digital tool aimed at organisation, planning, and prospective memory. Three major priorities were identified: emotional self-regulation, customisable supports for daily autonomy (routines and reminders with privacy controls), and practical psychoeducation for school and family, with significant feedback. These perspectives were collected through focus groups guided by a semi-structured script, chosen for its flexibility

and adaptation to the context, to facilitate the generation and exchange of ideas around the central theme, in line with previous approaches in health technology design [37]. Collaboration between developers, clinicians, educators, and families, along with the active participation of children and adolescents, is essential to create useful, adaptable, and motivating digital health solutions, focused on users' real needs, which can translate into improvements in daily functioning, social skills, participation, and quality of life [36].

The findings of this qualitative study reveal a complex but manageable set of needs, which serves as a guide for the design of a comprehensive digital intervention (emotional, functional, educational, and family) that goes beyond simple digitalisation. In this context, the focus groups made it possible to identify key issues related to content organisation, usability, motivation, reward systems, and personalisation [38].

In particular, the contributions of children and adolescents underline the centrality of executive functions as the core of their daily difficulties. The repeated need for personalised reminder systems, supports for daily planning, positive feedback, and tools that favour emotional and behavioural self-regulation is consistent with the literature that points to deficits in prospective memory, organisation, and emotional regulation in ADHD [12,39]. The mention of control and privacy mechanisms over reminders also indicates an explicit desire for autonomy and self-management, which reinforces the importance of avoiding paternalistic designs in digital interventions [40]. This type of technology has significant potential to offer immersive and motivating experiences, where children and adolescents can practice social skills and reduce anxiety in simulated social situations, favouring the generalisation of skills to real contexts [41]. Likewise, gamification—through rewards such as points, badges, or levels—has proven to be an effective strategy to capture and maintain attention, motivating children and adolescents to continue using these digital platforms and contributing to the learning of multiple skills [42]. Other authors have pointed out that digital interventions should include content about health care and self-management and be accessible to people with different care capacities. Components such as symptom monitoring, wearable devices, and SMS messaging can increase engagement [43]. It's important to do more research into practical tools, such as medication reminders and interactive assistants. Co-producing future solutions with stakeholders and adapting them to underserved populations will be key to moving towards more inclusive digital care for young people with ADHD [43].

Professionals involved in clinical and educational settings have shown interest in solutions that are easy to use, that allow the progress of children and adolescents to be monetised in real time, and that can be adapted to the different stages of development, promoting more effective and personalised interventions [44]. However, the perception and expectations regarding the usefulness of these technologies vary depending on previous experience, training, and willingness to incorporate digital solutions into their daily practices [45]. In the case of educational professionals, needs emerge that transcend direct help to students and point to a structural lack of specific training in emotional regulation strategies, academic planning, and classroom methodologies adapted to attentional difficulties. These findings coincide with studies that identify a training deficit in teachers regarding ADHD and a feeling of overflow due to the lack of specific resources [46]. The demand for evidence-based training resources, such as podcasts and up-to-date articles, reinforces the need to turn the digital tool into a platform not only for the student but for the entire educational community. Interventions using technologies such as robots, virtual reality, and interactive games create personalised and engaging experiences, as reported by educators and parents [44]. Also, when education professionals are involved in the intervention, they are more aware of their role in assisting students with special needs in school, and it is very important and effective for teachers, parents, educational psychologists, and other professionals to work together to provide a learning environment that is conducive to students with special needs rather than working separately [47].

For their part, occupational therapists emphasised a comprehensive approach, encompassing not only self-regulation but also activities of daily living and the transfer of functional learning to the home environment. This perspective, aimed at promoting the autonomy and motivation of children and adolescents, coincides with the growing evidence that advocates ecological and transversal interventions in real contexts [8]. The proposal to use technology as a motivational vehicle, while avoiding the abuse of the screen, underlines a key challenge in the design of digital interventions: maximising adherence without increasing technological dependence [48,49].

The analysis of the group of families clearly exposes the systemic impact of ADHD, especially in terms of parental overload, stress associated with daily supervision, and lack of understanding of the specific needs of their children. These concerns have been widely described in the literature, showing how ADHD not only affects the child, but family dynamics [50–52]. The explicit demand for psychoeducation content, both in audiovisual format and in self-help systems, points to a relevant opportunity for the digital tool as a space for family accompaniment [53], facilitating the improvement of parental self-efficacy and promoting healthier dynamics [54]. Parents may experience difficulties when trying to understand how devices work, generating the need for additional technical support [55]. In response, the incorporation of visual support materials—such as graphic or video tutorials—is recommended, as well as the availability of personalised technical assistance, especially for those users interested in integrating mHealth tools into behaviour-oriented interventions. In addition, consistent with other studies, concerns related to aspects such as usability, accessibility, privacy protection, and integration with traditional interventions also arose in the group of parents [56]. The implementation and design of these technologies must consider the particularities of each group, as well as the possible risks of excessive or misdirected use. User acceptance, ease of use, and integration of reward systems into everyday contexts are crucial elements to ensure the long-term sustainability and effectiveness of these digital interventions [57]. It is also advisable to anticipate difficulties associated with the use of devices—distraction, overuse of screens, digital divide, learning curve, or privacy issues—and address them by design through accessibility criteria and privacy principles.

Interpreting the findings through the WHO International Classification of Functioning, Disability and Health (ICF/ICF-CY) and the family of Participation-Related Constructs (fPRC) provided a coherent framework to situate stakeholder-identified needs within everyday participation contexts. Rather than focusing solely on individual impairments or skills, this perspective highlighted how participation challenges in children and adolescents with ADHD emerge from the dynamic interaction between functional abilities and environmental conditions across home, school, and community settings.

The identified themes mapped primarily onto ICF Environmental Factors, including products and technology, support and relationships, attitudes, and services and systems, underscoring the central role of contextual determinants in shaping participation. Framing the results using the fPRC further allowed differentiation between attendance and involvement, while incorporating related constructs such as activity competence, sense of self, and preferences. This integrative perspective supports a shift from symptom-focused approaches towards participation-oriented, context-responsive digital interventions.

In addition, the interpretation of the findings was informed by empirical literature on environmental determinants of participation, particularly studies using the Participation and Environment Measure for Children and Youth (PEM-CY) across home, school, and community contexts. This body of evidence provided a sensitising lens to support the labelling and interpretation of barriers and facilitators identified in the qualitative data. Across stakeholder groups, recurrent factors such as cognitive demands of tasks, sensory qualities of environments, availability of teacher and peer support, and organisational or policy-related features emerged as central determinants shaping participation for children and adolescents with ADHD. Situating these findings within established empirical frameworks strengthens their interpretive coherence and underscores the relevance of addressing contextual and environmental conditions when designing participation-oriented digital interventions.

The integration of perspectives across stakeholder groups revealed a coherent conceptual structure that informs future co-design processes by clarifying shared priorities, design considerations, and contextual requirements. Although each group emphasised different priorities—children focusing on reminders, autonomy, and emotional cues; parents highlighting family burden, psychoeducation, and communication; teachers emphasising classroom strategies and institutional constraints; and occupational therapists underscoring functional autonomy and motivation—these viewpoints converged around shared implementation determinants. When interpreted through CFIR domains (Intervention Characteristics, Individual Needs and Resources, Outer Setting, and Process) and fPRC constructs related to participation (attendance, involvement, activity competence, sense of self, preferences, and environmental fit), these diverse contributions aligned into a unified conceptual model. This model positions emotional regulation, routine-building supports, and psychoeducational resources as interdependent components embedded within an adaptable, context-responsive digital tool. The resulting framework reflects the environmental and relational dynamics that shape participation across home, school, and therapeutic settings and

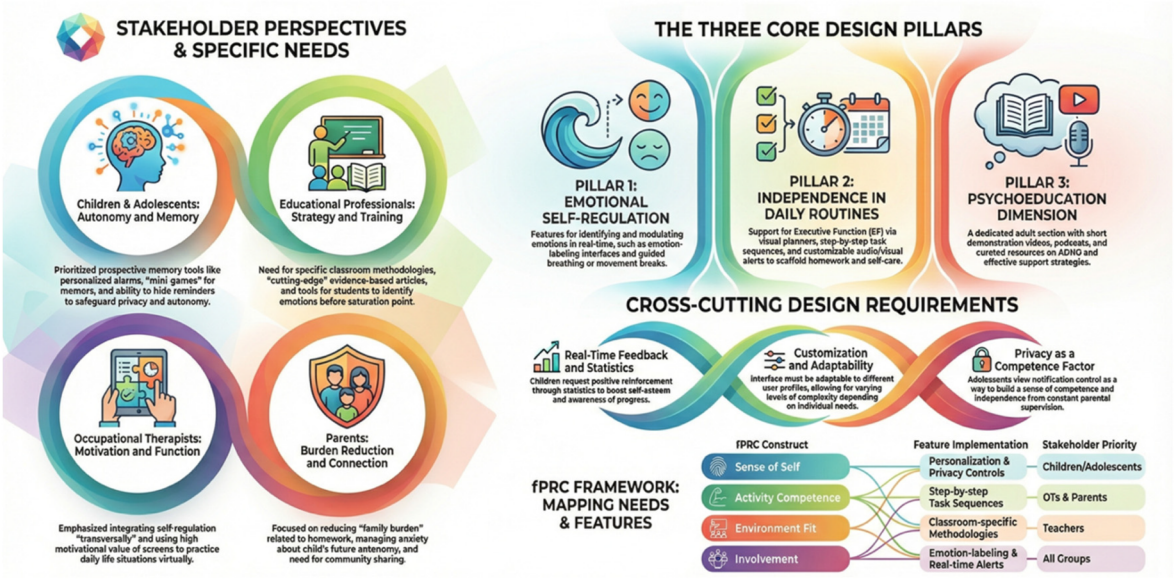


Figure 5. Integration of multi-stakeholder needs into a preliminary intervention model for ADHD.

provides a cohesive foundation for subsequent co-design phases [26,35,58–60]. This integration process is summarised in Figure 5.

Strengths and limitations

This study provides ecological value by integrating, through co-design, the perspectives of four key groups (children and adolescents, families, educational professionals, and occupational therapists), using a thematic guide anchored in the CFIR and a dual (deductive–inductive) analytic approach conducted by at least three independent researchers, which reinforces the credibility and traceability of the findings [34].

However, several limitations should be acknowledged. First, the sample was drawn from a single geographic region, which may limit the generalisability and transferability of the findings to other cultural or educational contexts. Second, the use of convenience and purposive sampling, while appropriate for qualitative research, may have introduced selection bias, as participants who were more motivated or had greater access to resources may be overrepresented. Third, although the number of focus groups aligns with established guidelines, formal assessment of data saturation was not conducted, which may mean that some relevant themes or minority perspectives were not fully captured.

Additionally, the study relied on self-reported data and group discussions, which may be influenced by social desirability bias or group dynamics, potentially limiting the depth or authenticity of some responses. The absence of longitudinal follow-up also precludes assessment of how stakeholder perspectives might evolve over time or after exposure to the digital intervention. Furthermore, while the inclusion of multiple stakeholder groups is a strength, the relatively small number of children and adolescents may have limited the diversity of experiences represented within this key group.

Finally, the findings are based on the early stages of needs assessment and do not include co-designing of the digital tool, which needs to be developed in future research, evaluating the impact and sustainability of the intervention in real-life contexts.

Conclusion

Together, the findings consolidate three functional blocks already described—(1) emotional/self-regulation, (2) functional (autonomy, planning, time management, and prospective memory), and (3) educational (understanding ADHD, stigma reduction, and evidence-based practices)—as axes of a user-centered design. Integrating these blocks into a customisable and modular platform aligns with current recommendations

for neurodevelopmental disorders (NDDs), where adaptability to the user profile is essential for efficacy and sustainability [61]. Finally, it is worth highlighting the coherence of the results with theoretical frameworks such as the Consolidated Framework for Implementation Research (CFIR) [34,37], showing that any digital tool must consider not only the needs of the individual (user characteristics), but also family dynamics (characteristics of the environment), the perceptions of professionals (characteristics of the intervention) and structural barriers in the educational and clinical context (characteristics of the system). The results allow us to propose future lines of development, highlighting the importance of co-designed, reiterated, and evidence-based strategies, as well as the evaluation of not only clinical results, but also daily functionality, quality of life, and satisfaction of the users involved [62].

Disclosure statement

No potential conflict of interest was reported by the author(s).

Author contributions

CRediT: **Marina Rivas-García**: Formal analysis, Writing – original draft, Writing – review & editing; **Carmen Vidal-Ramírez**: Data curation, Project administration, Writing – original draft, Writing – review & editing; **Marta Mora-Traverso**: Methodology, Writing – original draft, Writing – review & editing; **Abel Toledano-González**: Project administration, Writing – original draft, Writing – review & editing; **María del Carmen Rodríguez-Martínez**: Data curation, Writing – original draft, Writing – review & editing; **Esther Molina-Torres**: Project administration, Writing – review & editing; **José-Antonio Marín-Marín**: Data curation, Project administration, Writing – original draft, Writing – review & editing; **José-Matías Triviño-Juárez**: Data curation, Project administration, Writing – original draft, Writing – review & editing; **Miguel Gea-Mejías**: Conceptualization, Project administration, Software, Writing – review & editing; **Dulce Romero-Ayuso**: Conceptualization, Formal analysis, Funding acquisition, Project administration, Writing – original draft, Writing – review & editing.

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Notes on contributors

Marina Rivas-García, MD, works at the San Cecilio University Hospital in Granada and conducts research at the University of Granada. Her clinical and research work focuses on neurodevelopmental disorders, specifically ADHD, and on the implementation of digital health interventions to improve patient care and emotional regulation.

Carmen Vidal-Ramírez is a researcher at the University of Granada. Her work focuses on occupational science and neurorehabilitation, with a special emphasis on the development of assistive technologies and user-centered design to promote independence.

Marta Mora-Traverso is an occupational therapist and researcher. Her work explores the intersection between clinical practice and health education, with a special emphasis on reducing stigma and implementing evidence-based practices.

Abel Toledano-González is a professor and researcher at the University of Málaga. He specializes in the development of digital tools for clinical assessment. His experience encompasses the application of psychological theories to improve patient engagement and outcomes in digital health interventions.

María del Carmen Rodríguez-Martínez is a professor at the University of Málaga. Her research interests focus on the development of functional skills that contribute to the fields of occupational therapy and neurodevelopment.

Esther Molina-Torres is a school counselor and researcher at the University of Granada. Her work focuses on school counseling in early childhood and primary education, especially for children with behavioral disorders.

José-Antonio Marín-Marín is an academic at the University of Granada. He is an expert in educational technology and innovative teaching methodologies, and examines the impact of digital resources on the learning and inclusion of students with special educational needs.

José-Matías Triviño-Juárez is a researcher and clinician specializing in public health and primary care. His work focuses on integrating neurodevelopmental screening and support into healthcare systems, emphasizing multidisciplinary approaches to child development.

Miguel Gea-Mejías is an academic and researcher at the University of Granada, specializing in computer science. His work focuses on the technical architecture of digital health platforms, ensuring accessibility, modularity, and user-centered design for clinical applications.

Dulce Romero-Ayuso is a professor at the University of Granada and a leading researcher in neurodevelopmental disorders. Her research focuses on executive functions, emotional regulation, and the validation of digital health interventions, leading projects aimed at improving the autonomy and well being of neurodivergent individuals.

ORCID

Marina Rivas-García http://orcid.org/0009-0005-2705-0168
Carmen Vidal-Ramírez http://orcid.org/0000-0002-9856-671X
Marta Mora-Traverso http://orcid.org/0000-0003-0220-144X
Abel Toledano-González http://orcid.org/0000-0002-2496-0597
María del Carmen Rodríguez-Martínez http://orcid.org/0000-0002-0428-4798
Esther Molina-Torres http://orcid.org/0009-0005-9471-7989
José-Antonio Marín-Marín http://orcid.org/0000-0001-8623-4796
José-Matías Triviño-Juárez http://orcid.org/0000-0002-0988-3191
Miguel Gea-Mejías http://orcid.org/0000-0002-2603-3172
Dulce Romero-Ayuso http://orcid.org/0000-0002-2479-8913

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